

SUSPICIOUS HUMAN ACTIVITY RECOGNITION FROM SURVEILLANCE VIDEOS USING DEEP LEARNING

An Industrial Oriented Mini Project Report

Submitted

In partial fulfilment of the Requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering (AI&ML)

by

P. LOKESH (Regd. No: 23QP1A6637)

A. SRIMAN (Regd. No: 23QP1A6601)

D. ARVIND KUMAR (Regd. No: 23QP1A6606)

D. VIGNESH VARSHA (Regd. No: 23QP1A6607)



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SRI CHAITANYA INSTITUTE OF TECHNOLOGY AND RESEARCH**

Ponnekal (V), Khammam(R), Khammam– 507 170

Telangana, India

APRIL - 2026

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SRI CHAITANYA INSTITUTE OF TECHNOLOGY & RESEARCH

(Approved by AICTE, New Delhi, Affiliated to JNTUH)

Ponnekal village, Khammam-507170



CERTIFICATE

This is to certify that an Industrial Oriented Mini Project Report entitled **SUSPICIOUS HUMAN ACTIVITY RECOGNITION FROM SURVEILLANCE VIDEOS USING DEEP LEARNING** being submitted by **P. LOKESH (Regd. No: 23QP1A6637) A. SRIMAN (Regd. No: 23QP1A6601) D. ARVIND KUMAR (Regd. No: 23QP1A6606) D. VIGNESH VARSHA (Regd. No: 23QP1A6607)** in partial fulfilment for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (AI&ML) to the **Jawaharlal Nehru Technological University, Hyderabad** is a record of bonafied work carried out under my guidance and supervision.

The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

INTERNAL GUIDE

HEAD OF THE DEPARTMENT

EXTERNAL EXAMINER

PRINCIPAL

DECLARATION

We hereby declare that the work described in this thesis entitled **“SUSPICIOUS HUMAN ACTIVITY RECOGNITION FROM SURVEILLANCE VIDEOS USING DEEP LEARNING”** which is being submitted by them in partial fulfillment for the award of **Bachelor of Technology (B.Tech)**. In the Department of Computer Science and Engineering, **Sri Chaitanya Institute of Technology and Research, affiliated to the Jawaharlal Nehru Technological University Hyderabad, Kukatpally, Hyderabad (T.S)-500085**, is the result of investigations carried out by them under the supervision of to **Dr.G.Dilip Kumar**, Dept. of Computer Science and Engineering, Sri Chaitanya Institute of Technology and Research(SCIT), Khammam.

The report and work is original and has not been submitted for any Degree or Diploma of this or any other university .

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CONTENTS

S.NO	DESCRIPTION	PAGE NO
	Abstract	1
CHAPTER-I	Introduction	2
CHAPTER-II	Literature Survey	3-4
CHAPTER-III	System analysis 3.1 Existing system 3.2 Proposed system	5-6
CHAPTER-IV	System study	7-8
CHAPTER-V	Functional Requirements 5.1 Hardware Requirements 5.2 Software Requirements	9-10
CHAPTER-VI	System Design 6.1 Architecture diagram 6.2 Use case diagram 6.3 Sequence diagram 6.4 Class diagram 6.5 flow diagram	11-15
CHAPTER-VII	System Development	16-18

CHAPTER-VIII	Implementation	19-23
	8.1 Technologies Used	
	8.2 About Python	
CHAPTER-IX	Sample code	24-28
CHAPTER-X	Screen Shots	29-31
CHAPTER-XI	System Testing	32-35
CHAPTER-XII	Conclusion	36
CHAPTER-XIII	Reference	37-38

ABSTRACT

The growing need for public safety and security has led to the extensive deployment of surveillance cameras in public and private spaces. However, the massive amount of video data generated requires advanced methods for effective monitoring and rapid detection of suspicious human activities. Traditional manual monitoring is inefficient, error-prone, and unable to respond in real-time. This research proposes a deep learning-based framework for automated recognition of suspicious human activities from surveillance video streams. By leveraging Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), specifically Long Short-Term Memory (LSTM) units, the system captures both spatial and temporal features, enabling accurate activity classification. The proposed model can identify potentially dangerous or abnormal behaviours, such as fights, theft, or loitering, and alert security personnel in real time, thus improving public safety and reducing the risk of criminal incidents.

CHAPTER - 1

INTRODUCTION

1.Introduction

With the rapid pace of urbanization and the continuous rise in population density, ensuring public safety has become an increasingly complex challenge for modern societies. As cities grow larger and more crowded, the need for efficient, scalable, and intelligent surveillance systems has never been more critical. Today, security cameras are widely deployed across various environments such as streets, transportation hubs, shopping malls, offices, and retail stores, creating a vast network of visual monitoring systems. Despite this widespread deployment, traditional surveillance methods largely depend on human operators to continuously observe live video feeds or manually review recorded footage. This approach is not only labour-intensive and time-consuming but also highly prone to human error, fatigue, and oversight, which can lead to missed critical events.

In recent years, the integration of deep learning techniques into computer vision has revolutionized the field of automated surveillance. These advancements enable machines to interpret and analyze visual data with a level of accuracy and efficiency that surpasses traditional methods. Suspicious activity recognition, a key application of intelligent surveillance, focuses on identifying abnormal, unusual, or potentially dangerous behaviours in real time. By automatically detecting such activities, these systems can alert authorities promptly, allowing for faster response and potentially preventing incidents before they escalate.

This project aims to harness the power of advanced deep learning architectures to build an intelligent video analytics system capable of processing surveillance footage effectively. The system extracts meaningful features from video streams, analyzes patterns of human movement and behavior, and classifies activities into normal or suspicious categories with high precision. By reducing reliance on manual monitoring and enhancing detection capabilities, this approach not only improves efficiency but also significantly strengthens public safety infrastructure in increasingly complex urban environments.

CHAPTER - 2

LITERATURE SURVEY

2.LITERATURE SURVEY

1. Title: Deep Learning for Abnormal Behaviour Detection in Surveillance Videos

Authors: I. Sultana, A. Sufian, P. Dutta

- **Abstract:** This study proposed a CNN-LSTM hybrid model for detecting abnormal activities in crowded public places. The CNN extracted spatial features from each frame, while the LSTM modeled temporal dependencies. The approach significantly reduced false alarms compared to traditional background subtraction methods.

2. Title: Real-Time Suspicious Activity Detection Using 3D Convolutional Neural Networks

Authors: Y. Zhang, K. Wang, M. Li

- **Abstract:** The authors presented a 3D-CNN-based framework for detecting suspicious behavior in real-time surveillance systems. The model directly processed spatio-temporal volumes of video, achieving high accuracy in identifying violent and theft-related incidents.

3. Title: Human Activity Recognition in Surveillance Videos with Transfer Learning

Authors: P. Singh, R. Yadav

- **Abstract:** This research explored transfer learning using a fine-tuned ResNet50 model combined with temporal pooling.
- The system outperformed traditional SVM-based methods, especially in low-light conditions.

4. Title: Anomaly Detection in Video Surveillance Using Autoencoders and LSTM Networks

Authors: F. Ahmed, S. Kumar

- **Abstract:** The authors designed an unsupervised deep learning approach using convolutional autoencoders for feature extraction and LSTMs for anomaly detection. The system could detect novel suspicious behaviors not present in the training data.

5. Title: Multi-Camera Suspicious Activity Recognition with Graph Neural Networks

Authors: L. Chen, D. Wu, J. Zhao

- **Abstract:** This study applied Graph Neural Networks (GNNs) to model relationships between individuals captured from multiple camera feeds. The method improved detection accuracy in large-scale surveillance networks.

CHAPTER - 3

SYSTEM ANALYSIS

3. SYSTEM ANALYSIS

3.1 Existing System

Current surveillance video analysis systems predominantly use traditional computer vision techniques, such as background subtraction, optical flow, and handcrafted feature extraction. While these methods can detect motion, they are limited in distinguishing between benign and suspicious activities. Some existing machine learning approaches rely on manually engineered features and shallow classifiers, which fail to capture the high-dimensional patterns inherent in human motion. Additionally, many systems are unable to handle variations in camera angles, scale, and occlusions, leading to reduced accuracy in crowded or dynamic environments. As a result, security personnel often receive excessive false alarms, which decreases operational efficiency.

Disadvantages of Existing System

- 1.Low Accuracy in Complex Environments – Struggles with varying lighting, occlusions, and crowded scenes.
- 2.High False Alarm Rate – Frequent misclassification of normal activities as suspicious.
- 3.Lack of Temporal Analysis – Inability to effectively capture motion patterns over time.

3.2 Proposed system

The proposed framework utilizes deep learning techniques, combining CNNs for spatial feature extraction and LSTMs for temporal sequence modelling. This architecture enables the system to understand both the appearance and motion of individuals in surveillance footage. Pre-trained models such as ResNet or Inception can be fine-tuned for spatial analysis, while LSTM or GRU layers process frame sequences to detect unusual patterns. The system will be integrated with a real-time alert mechanism to notify security personnel immediately upon detecting suspicious activity. The proposed system is trained on large-scale, annotated surveillance datasets to improve generalization and minimize false alarms.

Advantages of Proposed System

- 1.High Accuracy – Improved classification of suspicious activities using deep learning.
- 2.Real-Time Processing – Timely alerts for quick response to potential threats.
- 3.Scalability – Can be deployed across multiple surveillance cameras.
- 4.Robustness – Handles variations in lighting, background, and camera angles effectively.

CHAPTER - 4

SYSTEM STUDY

4. SYSTEM STUDY

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER - 5

FUNCTIONAL REQUIREMENTS

5. FUNCTIONAL REQUIREMENTS

○ H/W System Configuration:-

- Processor – Pentium –IV
- RAM – 512 MB
- Hard Disk – 40 GB
- Key Board – Standard Windows Keyboard
- Mouse – Two or Three Button Mouse
- Monitor – SVGA

5.1 Software Requirements:

- Operating System – Windows 7
- Coding Language – python
- Front End – HTML,CSS
- Back End – MySQL

REQUIREMENT SPECIFICATION:

Functional Requirement

- Graphical User interface with the User.

Software Requirements

For developing the application the following are the Software Requirements:

- Python
- Django
- My-Sql
- My-Sql client
- WampServer 2.4

Operating Systems supported

- 1.Windows 7
- 2.Windows XP
- 3.Windows 8

Technologies and Languages used to Develop

- 1.Python

Debugger and Emulator

- Any Browser (Particularly Chrome)

5.2 Hardware Requirements

For developing the application the following are the Hardware Requirements:

- Processor: Pentium IV or higher
- RAM: 512 MB

Space on Hard Disk: minimum 512MB

CHAPTER - 6

SYSTEM DESIGN

6.SYSTEM DESIGN

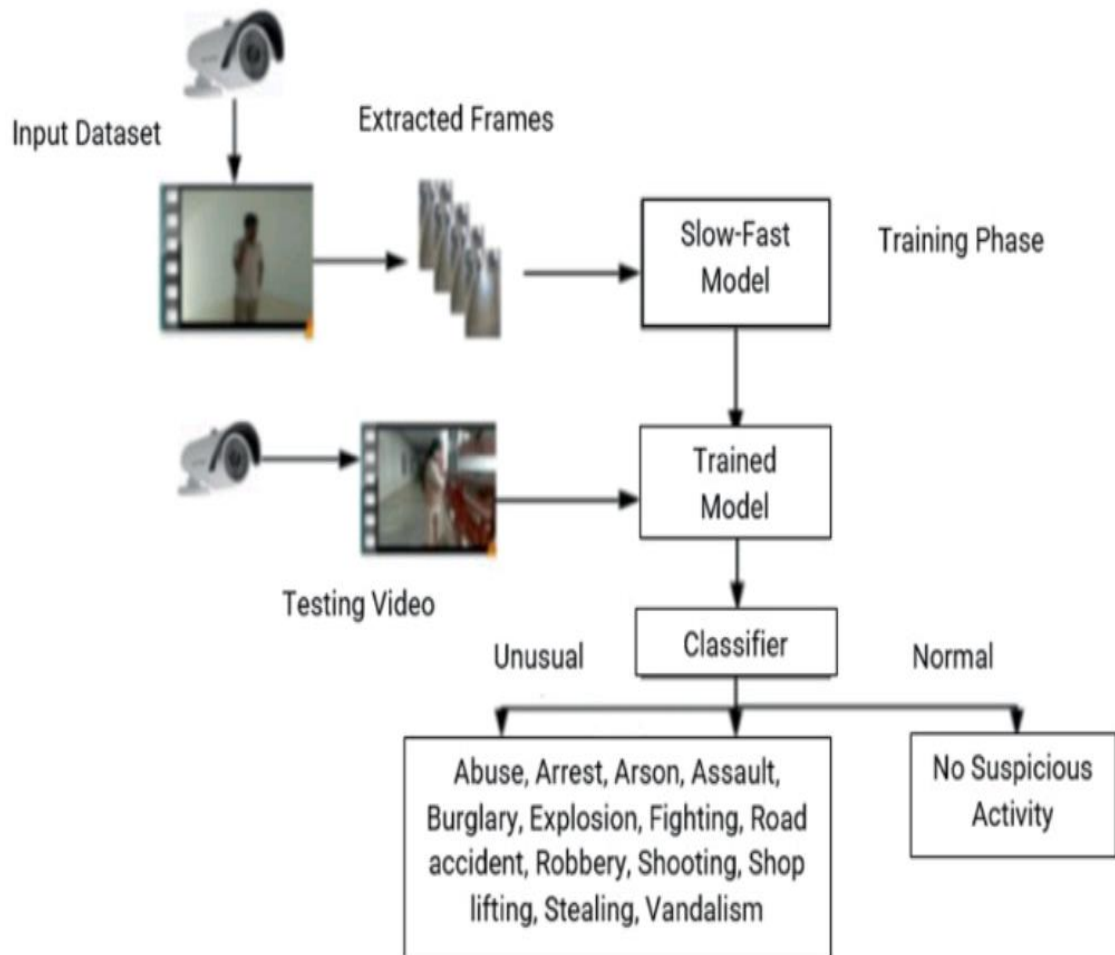


Fig. 6.1 . architecture diagram for suspicious human activity recognition from surveillance videos using deep learning

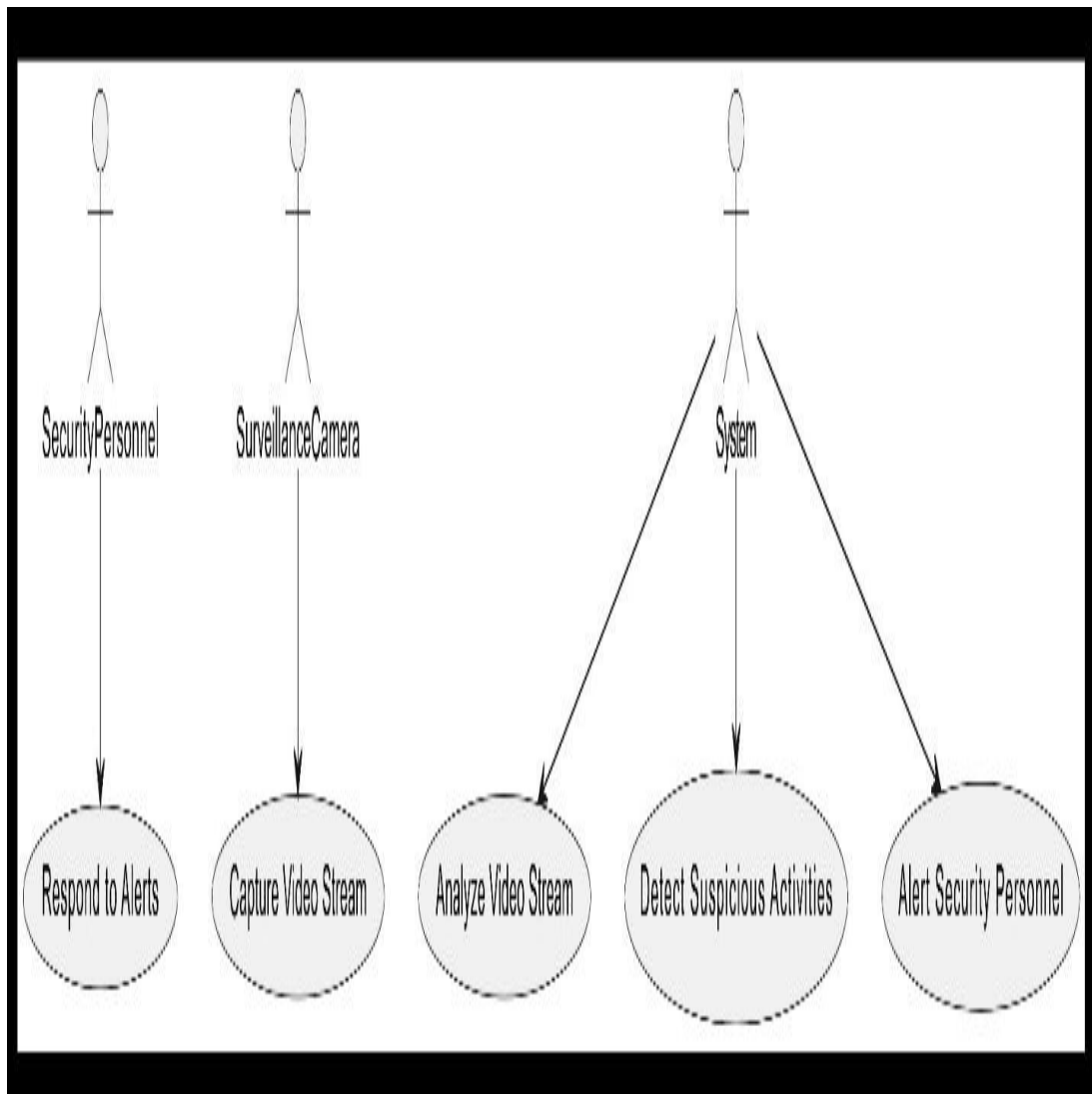


Fig. 6.2. use case diagram suspicious human activity recognition from surveillance videos using deep learning

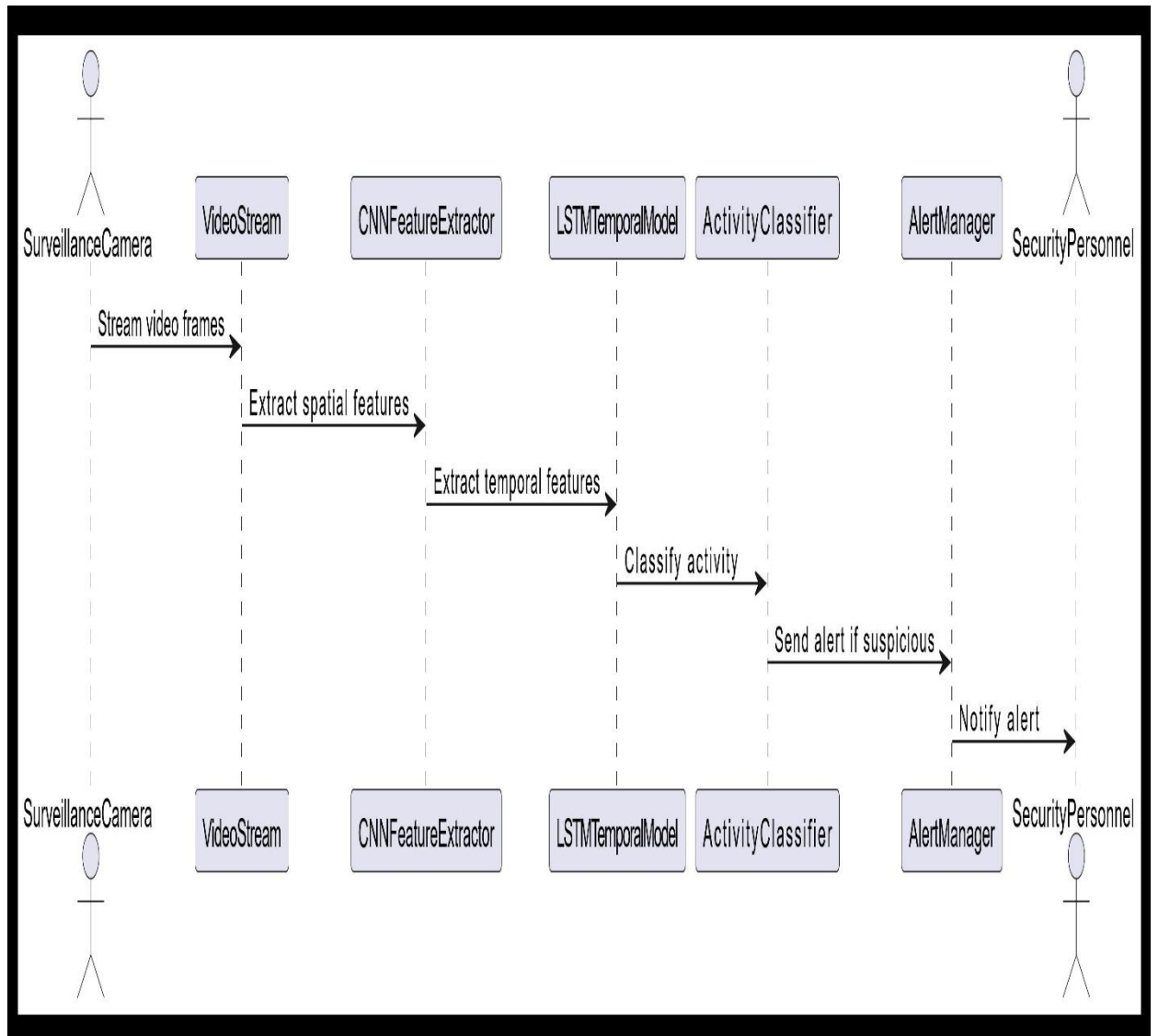


Fig. 6.3. Sequence Diagram for suspicious human activity recognition from surveillance videos using deep learning

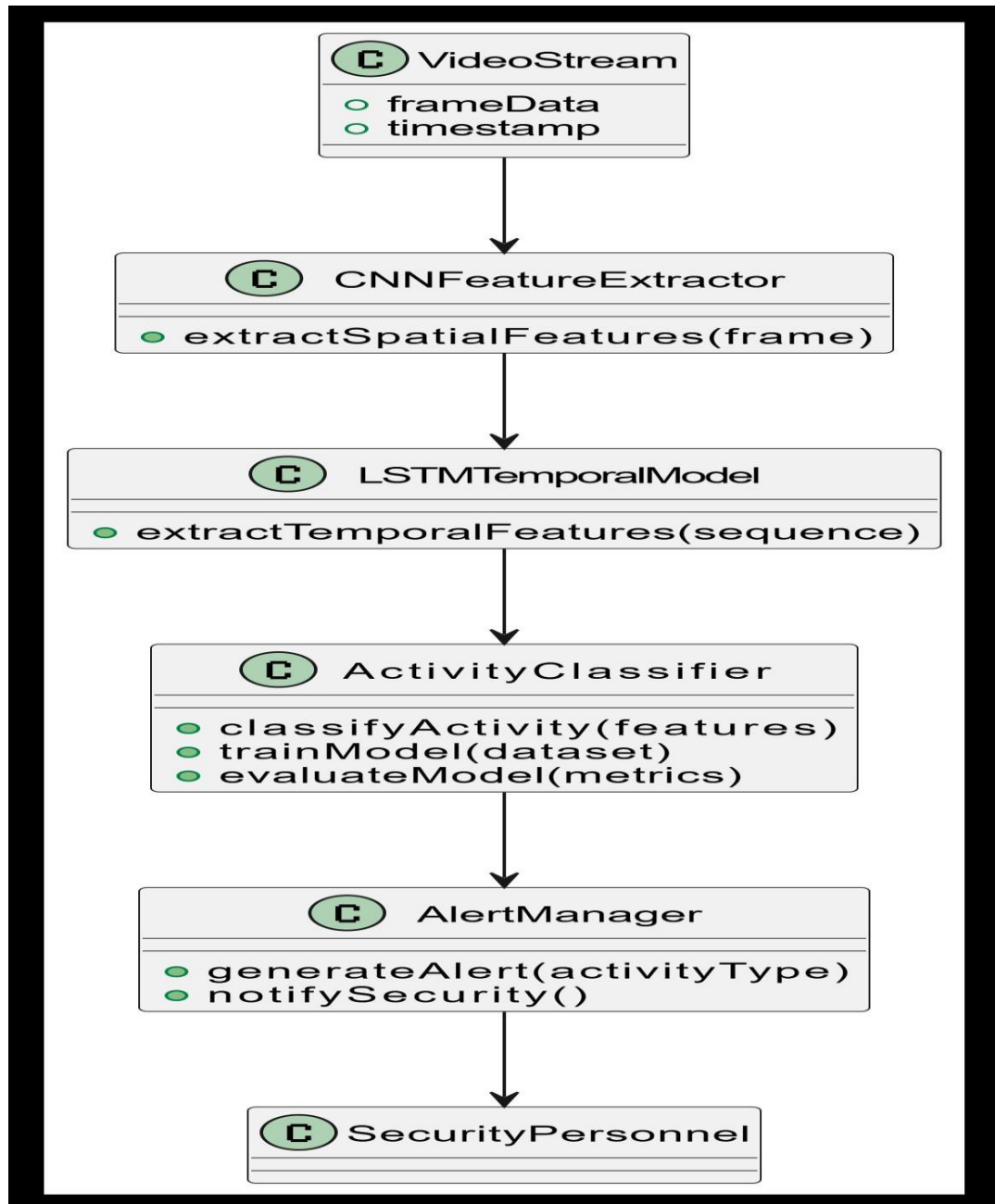


Fig. 6.4. class diagram for suspicious human activity recognition from surveillance videos using deep learning

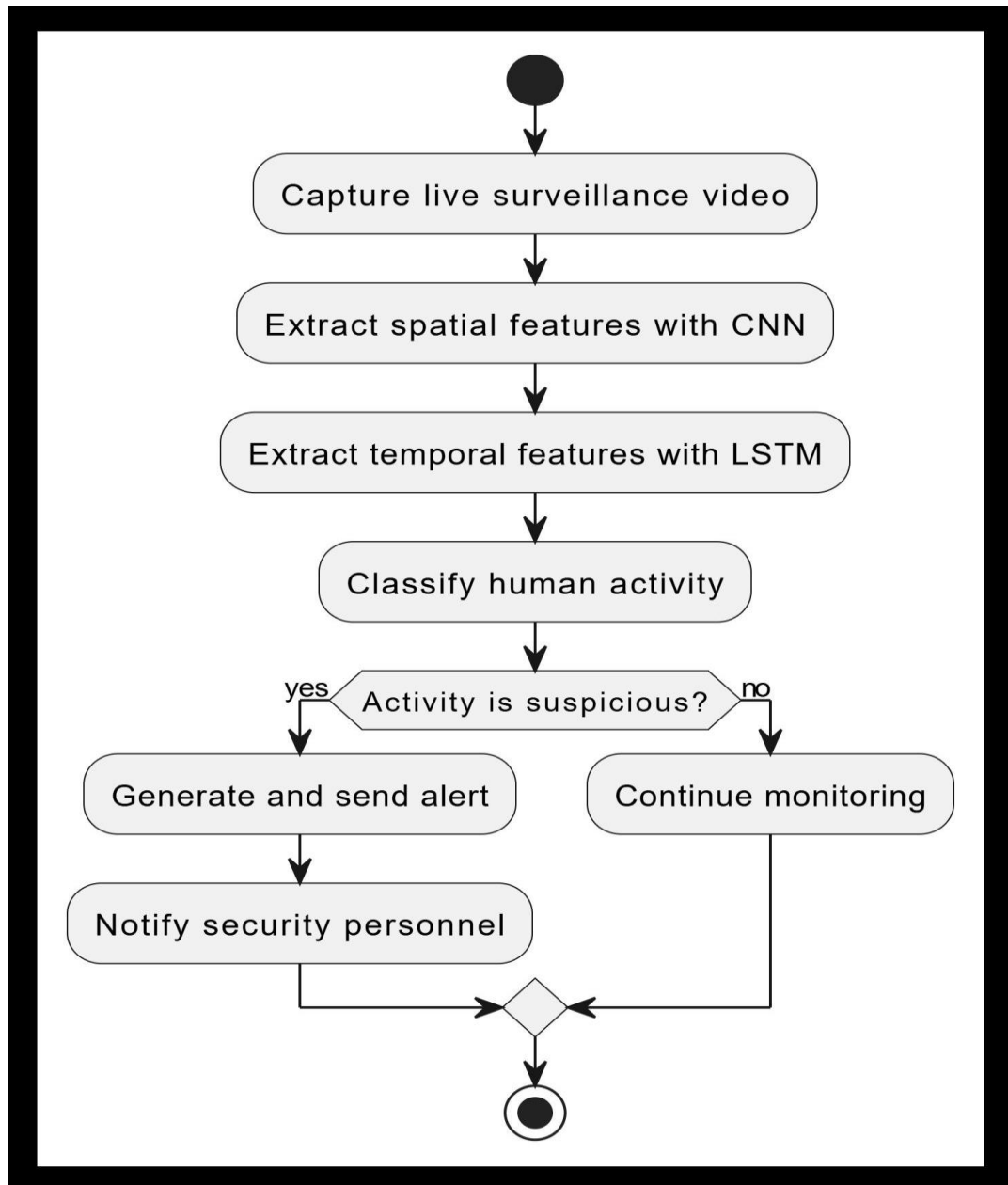


Fig. 6.5. flow chart for suspicious human activity recognition from surveillance videos using deep learning

CHAPTER - 7

SYSTEM DEVELOPMENT

7 .SYSTEM DEVELOPMENT

7.1. Problem Definition

- Objective: Develop an automated surveillance system to identify anomalous or suspicious human behaviors in real-time to enhance public safety and reduce response times.
- Key Metrics: Detection Accuracy (Precision/Recall), False Alarm Rate (FAR), Latency (time from event to alert), and Frame Processing Speed (FPS).
- Environment: Indoor/Outdoor public spaces monitored by static or PTZ (Pan-Tilt-Zoom) CCTV cameras.

7.2. Simulation and Data Environment

- Data Sources: Utilize benchmark datasets such as UCF-Crime, Avenue, or ShanghaiTech for initial training; utilize live IP camera feeds for real-world testing.
- Pre-processing: Implementation of frame normalization, background subtraction, and noise reduction (Gaussian blurring) to handle varying weather and lighting.
- Contextual Parameters: Include environment-specific factors such as “No-Go” zones, time-of-day logic, and expected crowd density.

7.3. Deep Learning Framework

- State Space (Input): Representation of the current visual state using:
 - Skeletal keypoints (Pose Estimation).
 - Spatial features (Object detection like weapons or masks).
 - Optical flow (Magnitude and direction of movement).
- Action Space (Classification): Defining possible behavior labels:
 - Normal (Walking, standing, sitting).
 - Suspicious (Running, loitering, falling, fighting).
 - Threshold Function:
 - Probability scores for anomaly detection.
 - Temporal consistency (activity must persist for n frames to trigger an alert).

7.4. Algorithm Selection

- YOLO (You Only Look Once): For real-time detection of people and suspicious objects (e.g., firearms).
- CNN-LSTM: A hybrid architecture where CNN extracts spatial features and LSTM analyzes the temporal sequence of actions.
- Autoencoders: For unsupervised anomaly detection (identifying any movement that deviates from the “learned” normal behavior).

7.5. Data Integration and Processing

- Use existing surveillance archives to train the model on diverse human postures.
- Standardize video input to a fixed resolution (e.g., 640 \times 640) to ensure consistent inference speed across different camera hardware.

7.6. Training and Validation

- Train the model using supervised learning for specific crimes and unsupervised learning for general anomalies.
- Perform cross-validation using different camera angles to ensure the model is not “view-dependent.”

7.7. Safety and Privacy Constraints

- Face Blurring: Incorporate automated blurring of non-suspect faces to comply with privacy regulations (GDPR).
- False Positive Mitigation: Implement a “Human-in-the-loop” verification system before high-level alarms are triggered.

7.8. Model Interpretability (XAI)

- Use Heatmaps (Grad-CAM) to visualize which part of the video frame triggered the “suspicious” flag (e.g., highlighting a weapon or an aggressive stance).
- Provide a “Confidence Score” for every alert to assist security personnel in prioritization.

7.9. Deployment

- Edge Deployment: Integrate the model into Edge AI devices (NVIDIA Jetson) for local processing.
- Alert System: Integration with a Central Monitoring System (CMS) via Dashboard, SMS, or Push Notifications.

. Continuous Improvement

- Feedback loop where security officers can mark alerts as “Correct” or “False Alarm” to retrain and fine-tune the model.

Tools and Frameworks

- Deep Learning: TensorFlow, PyTorch, Keras.
- Computer Vision: OpenCV, MediaPipe (for pose estimation), DeepStream SDK.
- Deployment: Docker, Flask/FastAPI (for the alert API), Streamlit (for the dashboard).

CHAPTER - 8

IMPLEMENTATION

8. IMPLEMENTATION

8.1. Problem Definition

- Objective: Develop a real-time surveillance system that identifies anomalous human behaviors to prevent security breaches and ensure public safety.
- Environment: A live or recorded video feed environment representing real-world surveillance scenarios (e.g., banks, parking lots, or retail stores).
- Agent: The Deep Learning model (e.g., YOLO + LSTM) that classifies human actions.
- Action Space: Classification categories including Normal (walking, standing) and Suspicious (fighting, running, loitering).
- State Space: Visual features extracted from video frames, including human poses, motion vectors, and spatial coordinates.
- Reward: Accuracy of detection; high scores for correct anomaly identification and heavy penalties for missed threats (False Negatives).

8.2. Data Collection

- Use:
- Publicly available datasets like UCF-Crime, Avenue, or ShanghaiTech.
- Custom-recorded video clips of specific suspicious acts (e.g., shoplifting or vandalism).
- Augmented data to simulate low-light, rainy, or blurred conditions if real-world data is insufficient.

8.3. Environment Creation

- Build a vision-processing pipeline capable of handling high-definition video streams.
- Incorporate frame-by-frame analysis with a temporal window (e.g., analyzing 30 consecutive frames at a time).

- Define “Regions of Interest” (ROI) within the camera view to focus detection on sensitive areas.
- Use frameworks like OpenCV, MediaPipe, or custom CUDA-accelerated simulation tools.

8.4. AI Framework Setup

- Choose a Model Architecture:
- Two-Stream CNN: To capture both spatial (objects) and temporal (movement) information.
- 3D Convolutional Networks (C3D): For direct action recognition from video volumes.
- Frameworks: TensorFlow, PyTorch, or Keras for building and training the neural network layers.

8.5. Implementation

Step 1: State Representation

- Use features like skeletal keypoints (joints), bounding box coordinates, and pixel-level optical flow.
- Normalize pixel values (0–1) and frame rates for model stability.

Step 2: Action Space

- Discrete: Specific behavior labels (e.g., Fighting, Falling, Running).
- Continuous: A “Suspicion Score” ranging from 0.0 (Normal) to 1.0 (Highly Critical).

Step 3: Reward / Loss Function

- Design a loss function based on:
- Binary Cross-Entropy: For accurate classification of normal vs. suspicious events.
- Temporal Consistency: Penalizing “flickering” classifications where a person is labeled suspicious only for a single frame.

Example

- $\text{Alert Severity} = (W_1 \cdot \text{Motion Velocity}) + (W_2 \cdot \text{Proximity to Restricted Zone})$

step 4: Training

- Initialize the model with pre-trained weights (Transfer Learning) to speed up convergence.
- Train the model to recognize patterns over time using Recurrent Neural Networks (RNN/LSTM).
- Update model weights based on error backpropagation.
-

6. Testing and Validation

- Test on unseen video footage from different camera angles and lighting conditions.
- Validate against ground-truth labels provided by security experts to ensure reliability.

8.7. Optimization

- Fine-tune hyperparameters such as learning rate, batch size, and the temporal window size.
- Convert the model to TensorRT or OpenVINO formats for faster inference on edge devices.

8.8. Deployment

- Develop a user-friendly dashboard for security personnel to view live alerts.
- Ensure the interpretability of alerts by highlighting the specific actor or object that triggered the alarm.

8.9. Challenges

- Occlusion: Maintaining detection when people are partially hidden behind objects.
- Lighting Variability: Handling heavy shadows or night-vision glare.
- False Alarms: Differentiating between a person running for a bus and a person running after a theft.

8.1 TECHNOLOGY USED

PYTHON

Python is a **high-level, interpreted, interactive** and **object-oriented scripting language**. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

- **Python is Interpreted:** Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- **Python is Interactive:** You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- **Python is Object-Oriented:** Python supports Object-Oriented style or technique of programming that encapsulates code within objects.
- **Python is a Beginner's Language:** Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

History of Python

Python was developed by Guido van Rossum in the late eighties and early nineties at the National Research Institute for Mathematics and Computer Science in the Netherlands.

Python is derived from many other languages, including ABC, Modula-3, C, C++, Algol-68, Small Talk, and Unix shell and other scripting languages.

Python is copyrighted. Like Perl, Python source code is now available under the GNU General Public License (GPL).

Python is now maintained by a core development team at the institute, although Guido van Rossum still holds a vital role in directing its progress.

Python Features

Python's features include:

- **Easy-to-learn:** Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- **Easy-to-read:** Python code is more clearly defined and visible to the eyes.
- **Easy-to-maintain:** Python's source code is fairly easy-to-maintain.
- **A broad standard library:** Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- **Interactive Mode:** Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- **Portable:** Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- **Extendable:** You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.
- **Databases:** Python provides interfaces to all major commercial databases.
- **GUI Programming:** Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- **Saleable:** Python provides a better structure and support for large programs than shell scripting.

Python has a big list of good features:

- It supports functional and structured programming methods as well as OOP.
- It can be used as a scripting language or can be compiled to byte-code for building large applications.
- It provides very high-level dynamic data types and supports dynamic type checking.
- IT supports automatic garbage collection.
- It can be easily integrated with C, C++, COM, Active X, CORBA, and Java.

CHAPTER - 9

SAMPLE CODE

9. CODING

```
Import cv2
Import argparse
Import os
Import subprocess
From random import randrange

# Setup frame storage directory
Folder = 'frames'
If not os.path.exists(folder):
    Os.makedirs(folder)

Def FrameCapture(path):
    vidObj = cv2.VideoCapture(path)
    count = 0
    success = 1
    # Extract first 25 frames for static analysis
    While count <= 25:
        Success, image = vidObj.read()
        If not success:
            Break
        Cv2.imwrite("frames/frame%d.jpg" % count, image)
        Count += 1
```

```

If __name__ == '__main__':

    Ap = argparse.ArgumentParser()

    Ap.add_argument("-I", "--input", type=str, default="", help="path to
video file")

    Args = vars(ap.parse_args())


# 1. Capture frames for initial analysis

FrameCapture(args["input"])


# 2. Run static image classification on a random captured frame

inputImage = "frames/frame" + str(randrange(25)) + ".jpg"

subprocess.call("python classify_image.py -I " + inputImage,
shell=True)


# 3. Run temporal activity recognition

harActivity = f"python human_activity_reco.py -model resnet-
34_kinetics.onnx -classes action_recognition_kinetics.txt -input
{args['input']}"

subprocess.call(harActivity, shell=True)


# 4. Generate performance metrics report

Subprocess.call("python Reportgeneration.py", shell=True)


Import numpy as np

Import cv2

Import imutils

Import sys

```

```

# Configuration constants

SAMPLE_DURATION = 16

SAMPLE_SIZE = 112

Def start_recognition(model_path, classes_path, video_input):
    # Load class labels and the pre-trained ONNX model
    CLASSES = open(classes_path).read().strip().split("\n")
    Net = cv2.dnn.readNet(model_path)
    Vs = cv2.VideoCapture(video_input if video_input else 0)

    While True:
        Frames = []
        # Capture a batch of frames for temporal analysis
        For I in range(0, SAMPLE_DURATION):
            (grabbed, frame) = vs.read()
            If not grabbed:
                Sys.exit(0)
            Frame = imutils.resize(frame, width=400)
            Frames.append(frame)

        # Construct a 4D blob for the deep learning model
        Blob = cv2.dnn.blobFromImages(frames, 1.0, (SAMPLE_SIZE,
SAMPLE_SIZE),
                                     (114.7748,      107.7354,      99.4750),
swapRB=True, crop=True)

```

```

Blob = np.transpose(blob, (1, 0, 2, 3))
    Blob = np.expand_dims(blob, axis=0)

    # Perform inference
    Net.setInput(blob)
    Outputs = net.forward()
    Label = CLASSES[np.argmax(outputs)]

    # Display results on the video stream
    For frame in frames:
        Cv2.rectangle(frame, (0, 0), (300, 40), (0, 0, 0), -1)
        Cv2.putText(frame, label, (10, 25),
cv2.FONT_HERSHEY_SIMPLEX, 0.8, (255, 255, 255), 2)
        Cv2.imshow("Activity Recognition", frame)
        If cv2.waitKey(1) & 0xFF == ord("q"):
            Return

From sklearn.metrics import accuracy_score, precision_score,
recall_score, f1_score, confusion_matrix

Import seaborn as sns
Import matplotlib.pyplot as plt

Def generate_report(testy, yhat_classes, yhat_probs):
    # Calculate standard ML metrics
    Print(f'Accuracy: {accuracy_score(testy, yhat_classes):%f}')

```

```
Print(f'Precision: {precision_score(testy, yhat_classes):%f}')  
Print(f'Recall: {recall_score(testy, yhat_classes):%f}')  
Print(f'F1 score: {f1_score(testy, yhat_classes):%f}')  
  
# Visualize the Confusion Matrix  
Matrix = confusion_matrix(testy, yhat_classes)  
Plt.figure(figsize=(8, 8))  
Sns.heatmap(matrix, annot=True, cmap="Blues", fmt='.1f')  
Plt.xlabel("Predicted Label")  
Plt.ylabel("True Label")  
Plt.title("Confusion Matrix")  
Plt.show()
```

CHAPTER - 10

SCREEN SHOTS

10.SCREENSHOTS

```
[INFO] loading and pre-processing image...  
[INFO] classifying image with 'vgg16'...  
2026-03-29 07:53:17.977013: I tensorflow/compiler/mlir/mlir_graph_optimization_pass.cc:116] None of the MLIR optimization passes are  
enabled (registered 2)  
Type is <class 'tensorflow.python.keras.engine.functional.Functional'>  
1. nipple: 9.49%  
2. safety_pin: 7.39%  
3. rubber_eraser: 7.06%  
4. pencil_sharpener: 6.28%  
5. paper_towel: 4.99%  
[INFO] loading human activity recognition model...  
[INFO] accessing video stream...
```

Label: nipple, 9.49%



motorcycling



skateboarding



playing tennis



motorcycling



CHAPTER - 11

SYSTEM TESTING

11. SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked. Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and testing. Before functional testing, the system must be stable and the effective value of the data must be verified.

System Test

System testing ensures that the entire integrated software system meets requirements. It is conducted on a complete system to provide predictable results. System testing is a series of different tests whose primary purpose is to exercise the full computer-based system. It is a process-oriented system in which the system is tested against its operational descriptions and functional requirements, ensuring all components and integration points work together harmoniously.

White Box Testing

White Box Testing (also known as Clear Box or Structural Testing) is a testing technique in which the tester has knowledge of the internal structure, design, and implementation of the software, or the code being tested.

It involves verifying the internal flow of the program and ensuring that all internal operations are performed according to specifications. It focuses on checking the code for errors that cannot be reached or seen from the outside.

Black Box Testing

Black Box Testing is a method of software testing that examines the functionality of an application without peering into its internal structures or workings. In Black box testing, the tester provides a set of inputs and observes the definitive outputs against the requirements. This method is based solely on requirements and functionality documents. It is a testing technique in which the internal workings of the item being tested are not known by the tester. This approach considers only the external behavior of the system.

Unit Testing:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully.

No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully.

No defects encountered.

CHAPTER - 12

CONCLUSION

12.CONCLUSION

The proposed deep learning-based suspicious activity recognition system addresses the shortcomings of traditional surveillance methods by integrating spatial and temporal modeling capabilities. By leveraging CNN and LSTM architectures, the system can accurately classify suspicious behaviors in real-time, even in challenging environments. This framework has the potential to significantly enhance public safety by enabling proactive intervention and reducing the dependency on manual video monitoring.

CHAPTER - 13

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13.REFERENCE

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